



# ■ CS177 Bioinformatics: Software Development and Applications

## Lecture 22: Synthetic Biology

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# Outline



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- Overview of synthetic biology
  - Definition, scope, applications
  - Roadmap (the DBTL cycle)
- Engineering DNA
- Biomolecular engineering
- Host engineering (to skip here)
- Data science

Slides adapted from EBRC (Engineering Biology Research Consortium):

<https://ebrc.org/wp-content/uploads/2022/09/Intro-to-Synthetic-Biology-1.pdf>

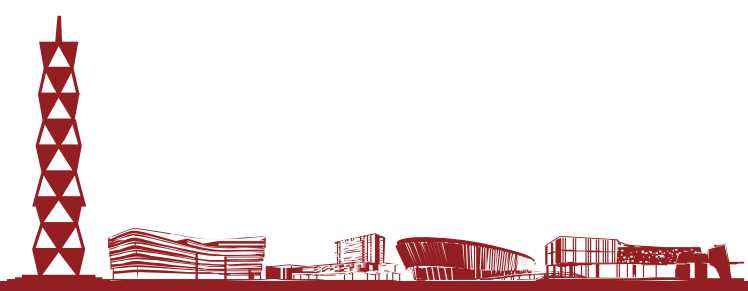


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# ■ What is Synthetic Biology?

- **Engineering Biology** (a broader term for Synthetic Biology) is the **design and construction of new biological entities**, such as:
  - Enzymes
  - Genetic circuits
  - Cells
  - The redesign of existing biological systems
- Building on the advances in molecular, cell, and systems biology, Engineering Biology **seeks to transform biology** in the same way that:
  - Synthesis transformed chemistry
  - Integrated circuit (IC) design transformed computing





# What make engineering biology unique?

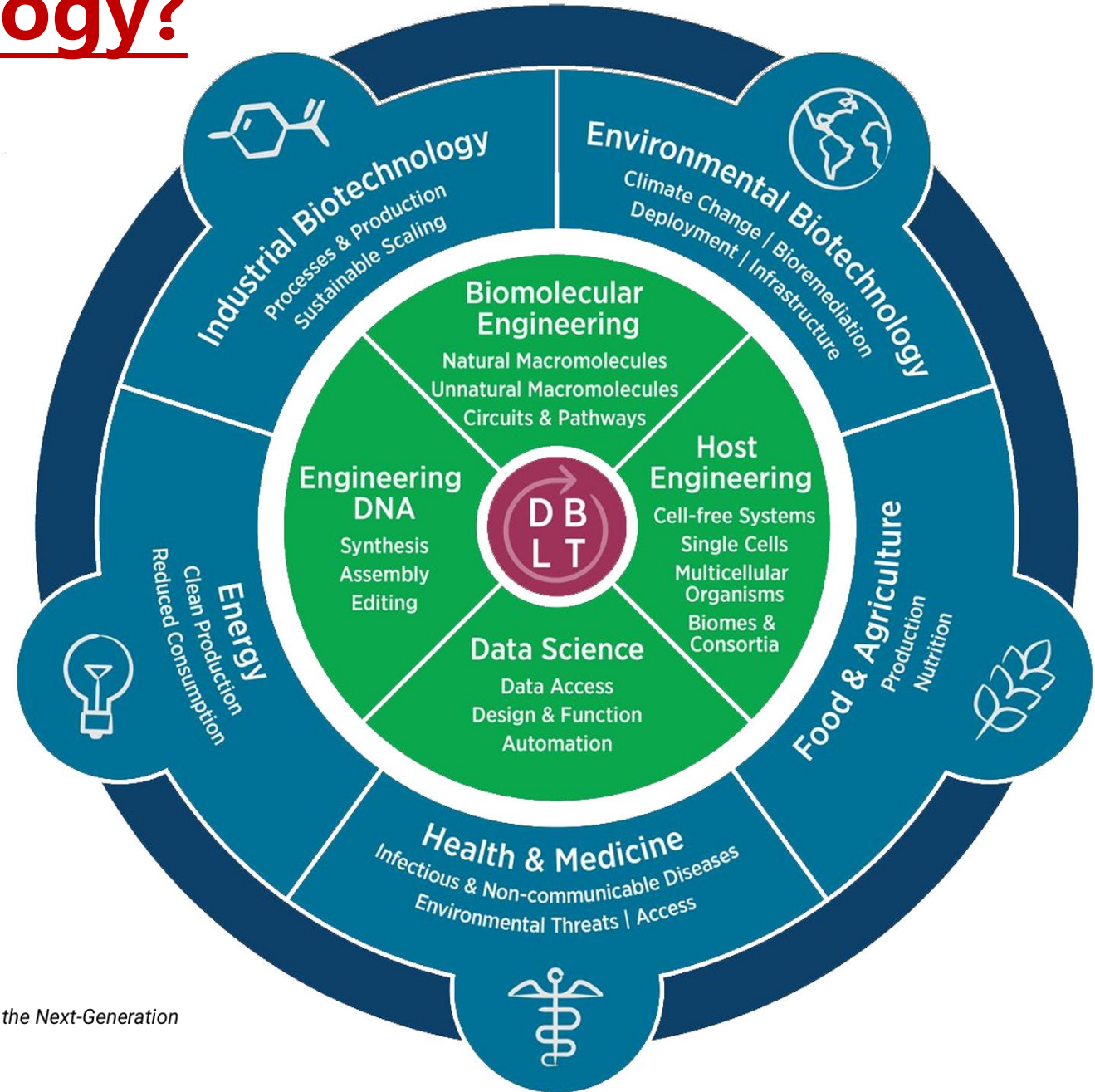


- The element that distinguishes engineering biology from traditional molecular and cellular biology is the **focus on the design and construction of core components**:
  - E.g. parts of enzymes, genetic circuits, metabolic pathways
  - They can be modeled, understood, and tuned to meet specific performance criteria
  - And the assembly of these smaller parts and devices into larger integrated systems to solve specific problems
- Unlike many other areas of engineering, biology is incredibly **non-linear and less predictable**, and there is **less knowledge of the parts** and **how they interact**
- Hence, the overwhelming physical details of natural biology (gene sequences, protein properties, biological systems) must be organized and recast via a set of design rules
  - Hide information
  - Manage complexity
  - Enable the engineering of many-component integrated biological systems



# What is Synthetic Biology?

The goal: To “engineer living cells to do something useful; for example, treat a disease, sense a toxic compound in the environment, or produce a valuable drug.” (BioBuilder)

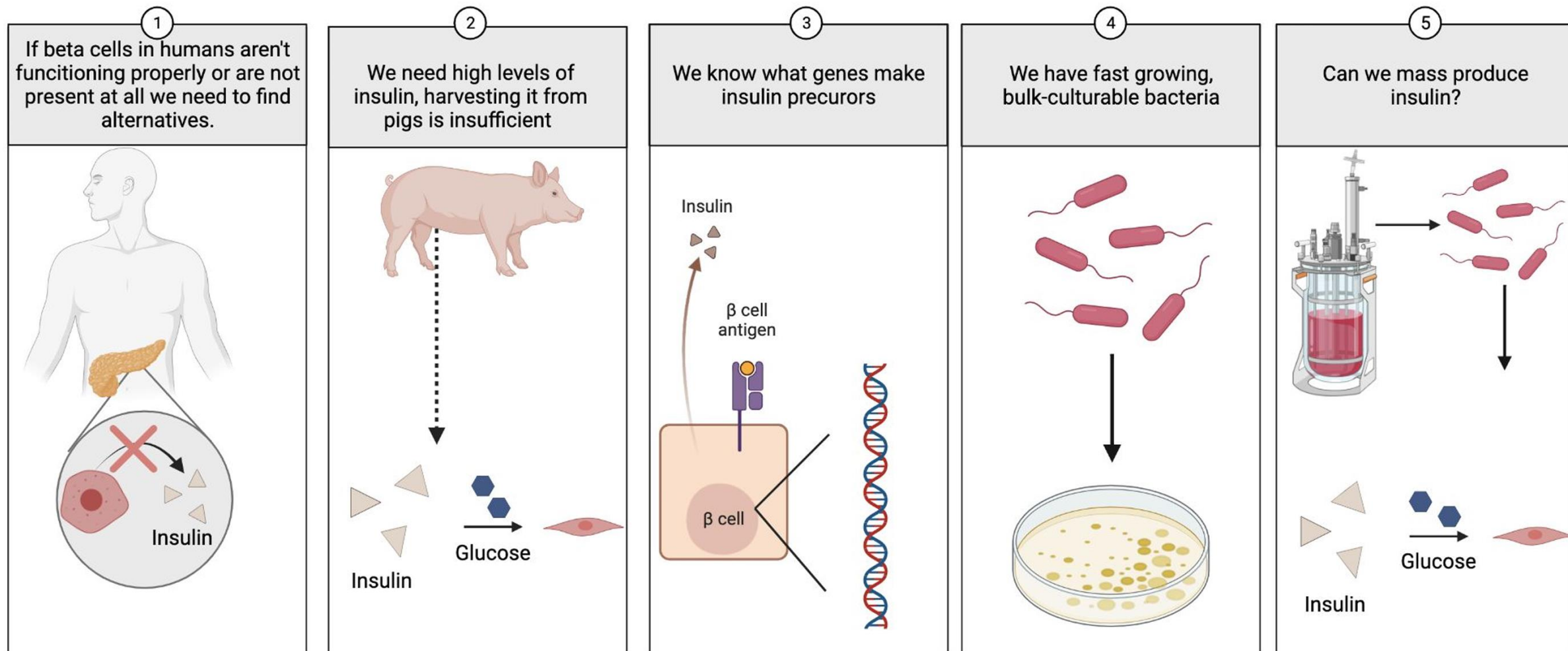




# Industrial: Bacterial production of insulin



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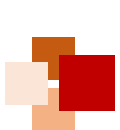


Mona Dabbas & Javin Oza, California Polytechnic State University.



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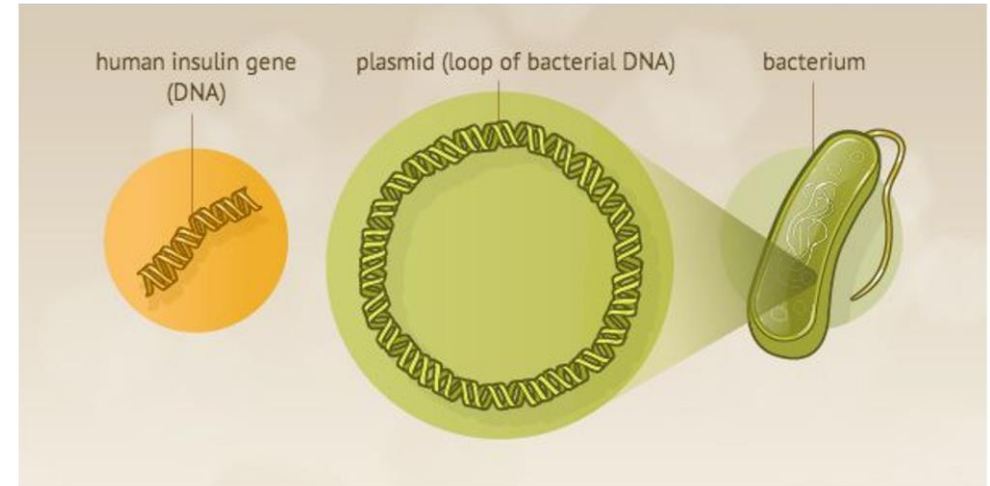
# Industrial: Bacterial production of insulin



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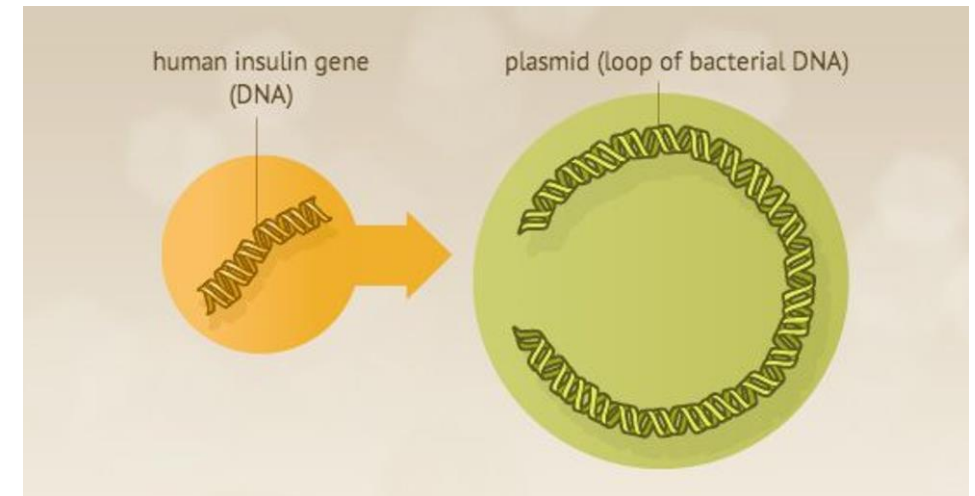
What tools do we need?

- Insulin precursor genes
- Genetic parts for gene expression in *E. coli*
- Way to assemble these genetic parts
- The *E. coli* cells themselves
- Way to get the DNA into the cells
- Way to harvest insulin from bacteria
- Way to measure insulin production



What steps do we need to take?

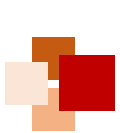
- Design a genetic circuit that will produce insulin in *E. coli*
- Build the genetic circuit, and put it into bacteria
- Test how much insulin is made
- Learn what did/didn't work, change design accordingly



NIH. From DNA to Beer: Harnessing Nature in Medicine & Industry. 2013



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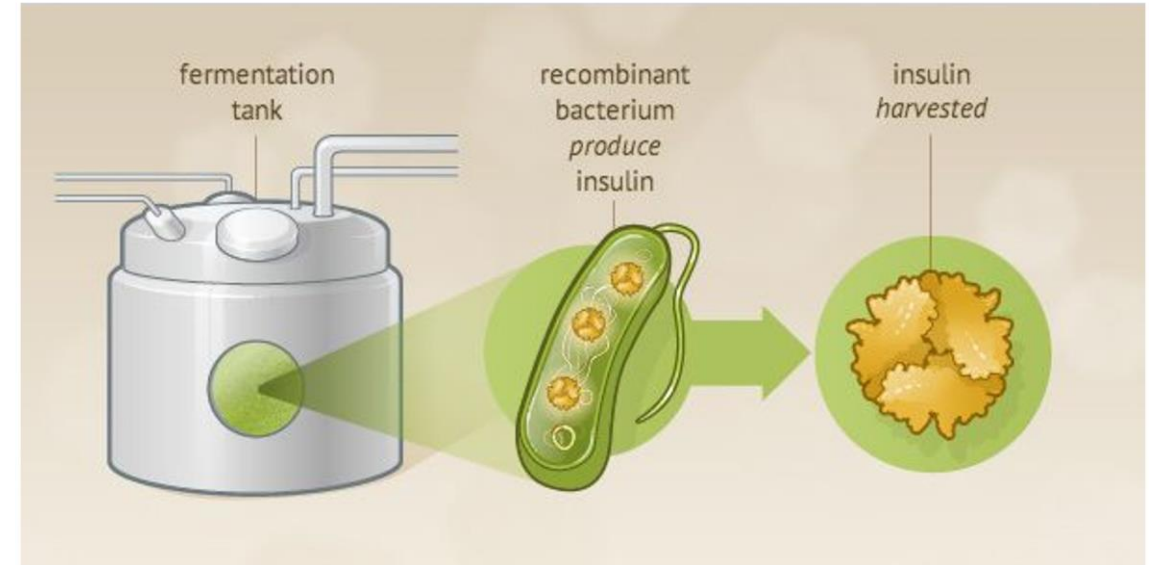
# Industrial: Bacterial production of insulin



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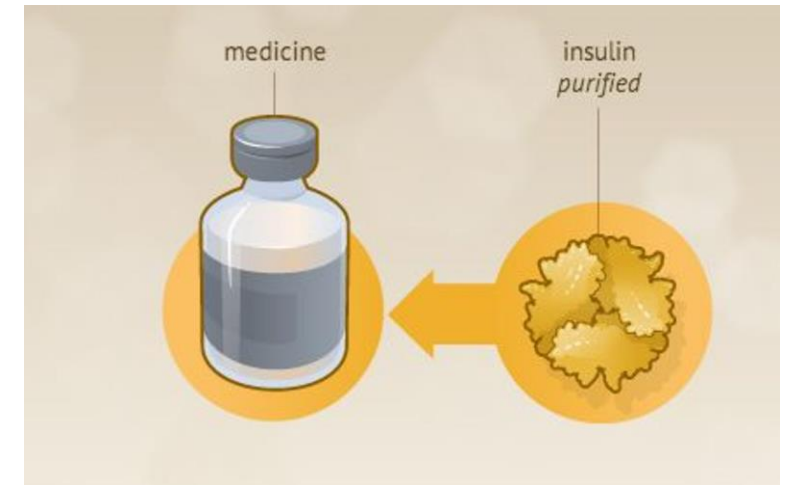
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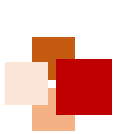


NIH. From DNA to Beer: Harnessing Nature in Medicine & Industry. 2013



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# SynBio Roadmap

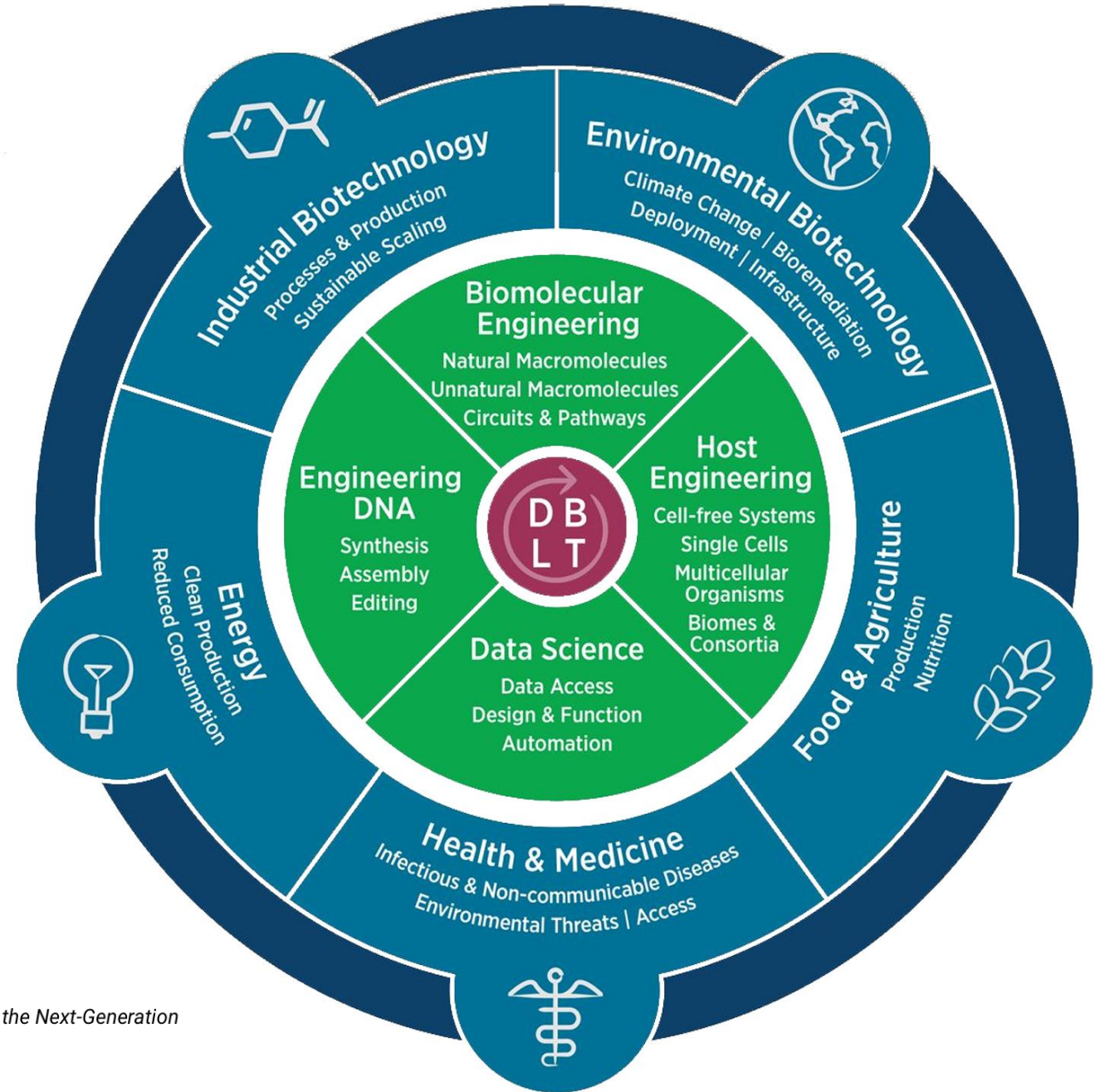


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From a Design, Build, Test, Learn  
centered approach

Uses biological, engineering, and  
computational tools

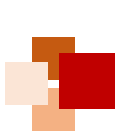
To positively impact multiple sectors



Engineering Biology Research Consortium (2019). *Engineering Biology: A Research Roadmap for the Next-Generation Bioeconomy*. Retrieved from <http://roadmap.ebrc.org>. doi: 10.25498/E4159B.



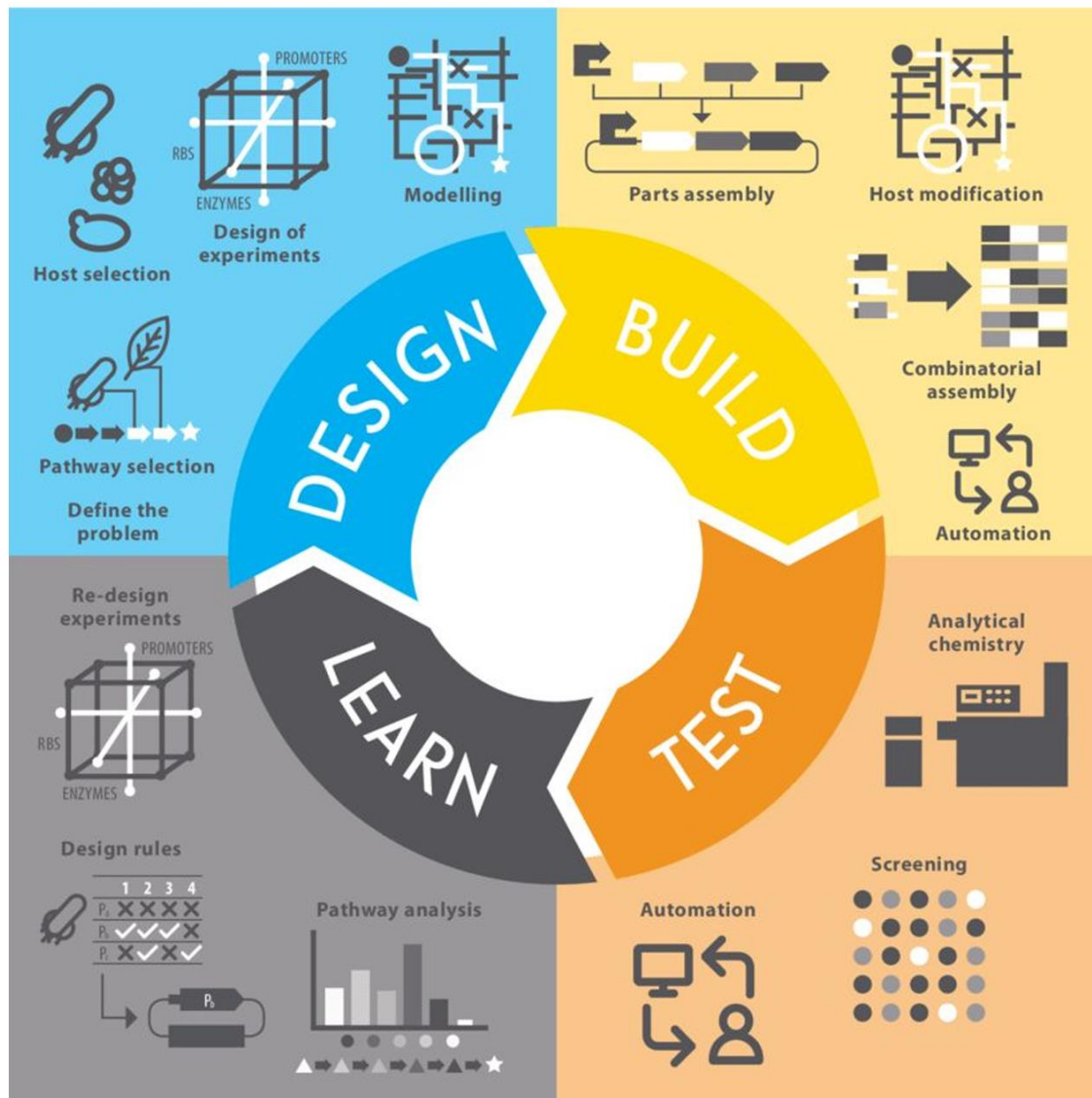
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# The DBTL Cycle



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**Design** a biological circuit or system for a specific function using literature & computational tools.

**Build** your system by assembling, editing, and installing genetic parts.

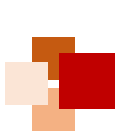
**Test** system functionality using robust metrics & controls.

**Learn** from how the system (doesn't) work to create new design rules, and share your knowledge.

Gray, P. et al. *Synthetic Biology in Australia: An Outlook to 2030* (Australian Council of Learned Academies, 2018).



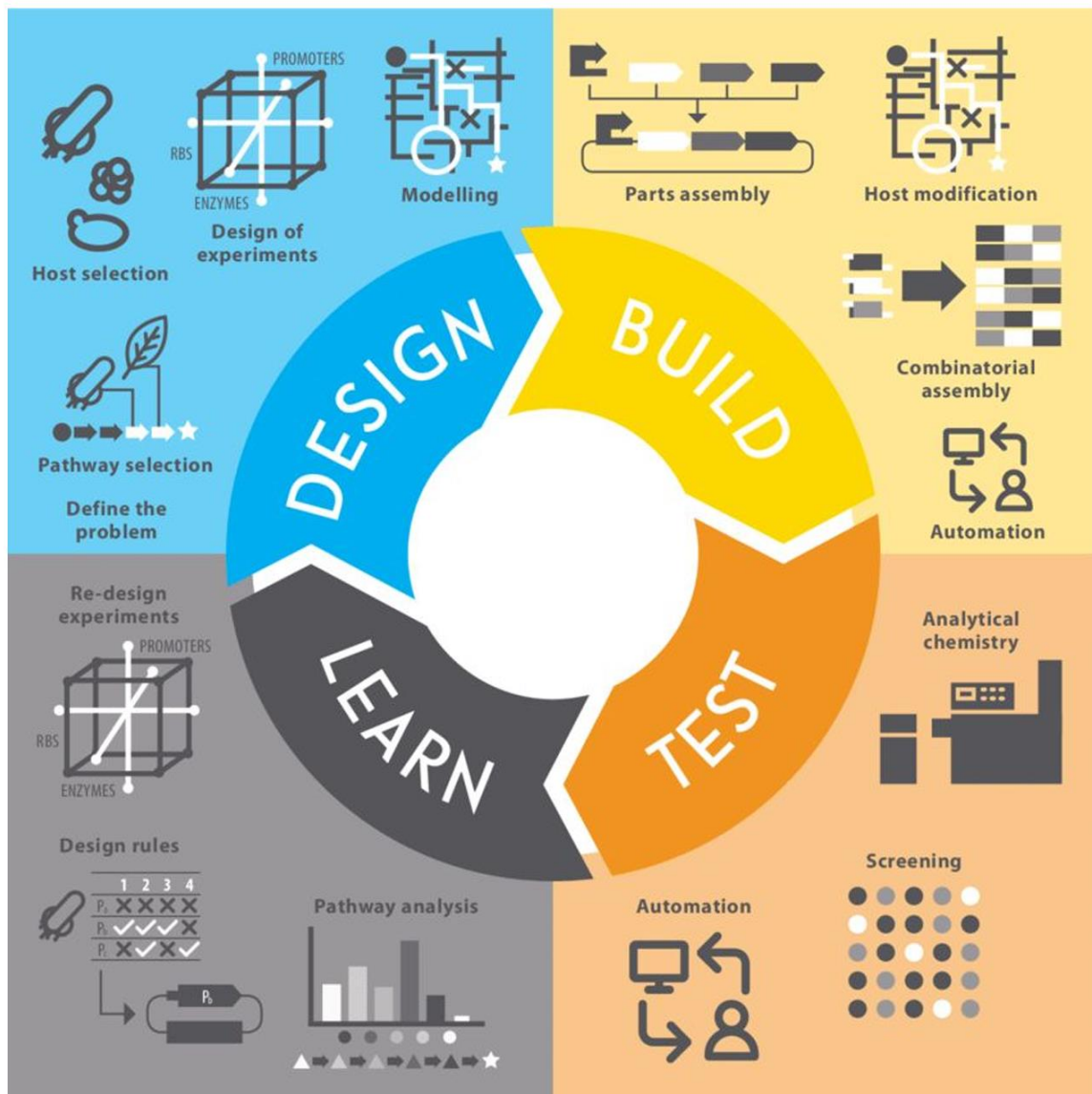




# DBTL: In Practice



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**Design** by finding genes for human insulin precursors, and optimize/model *E. coli* expression.

**Build** gene sequence including insulin precursor genes through synthesis & assembly.

**Test** your circuit in cells by measuring for insulin precursor proteins chemically via an assay.

**Learn** how well your system works, and what optimization is necessary.

Gray, P. et al. *Synthetic Biology in Australia: An Outlook to 2030* (Australian Council of Learned Academies, 2018).



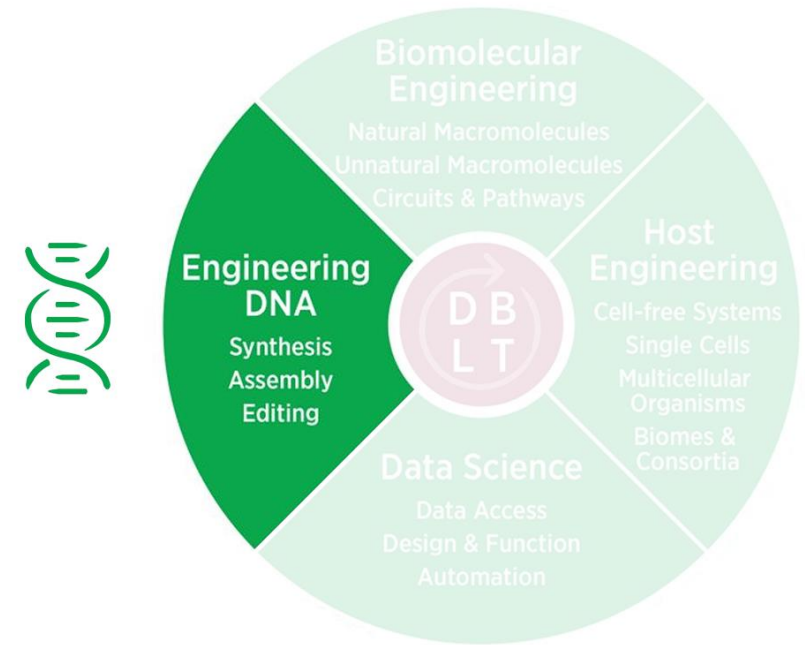


# Core tools for Engineering DNA



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- What are the current tools for engineering DNA?
- Tools enabling a engineering DNA DBTL
  - Synthesis
  - Sequencing
  - Standardization
  - Assembly
  - Editing



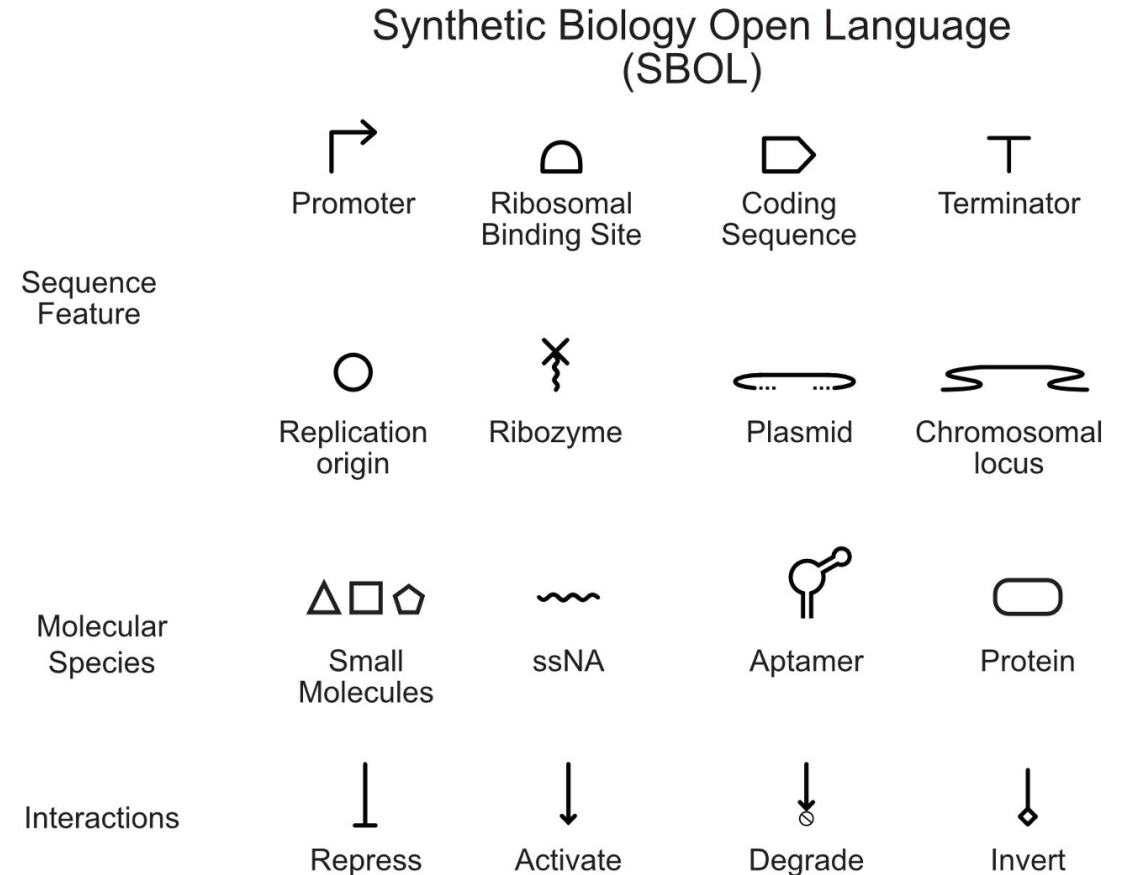
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# Engineering DNA: Standardized depiction of gene circuits



- Engineering Biology often requires descriptions of the DNA sequences used and how they function
- With the wealth of easy to synthesize DNA parts how do we communicate what pieces of synthetic DNA do?
- The [Synthetic Biology Open Language](#) (SBOL) is a framework for describing synthetic DNA sequences and their functional relationships

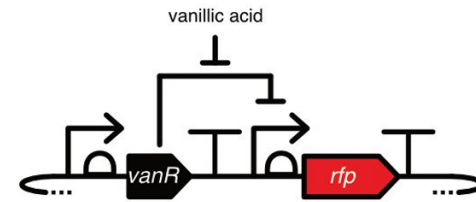




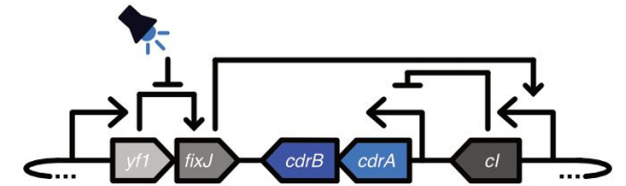
# Engineering DNA: Examples of SBOL-based diagrams



- On the left is a plasmid for chemically-inducible expression red fluorescent protein
- On the right is a plasmid for the light-inducible expression of the adhesins CdrAB
- Both of these plasmids express components that control the expression of other components
- The logical interpretations of the diagrams are listed below the image.



```
if(vanillic acid) :  
  express(rfp)
```

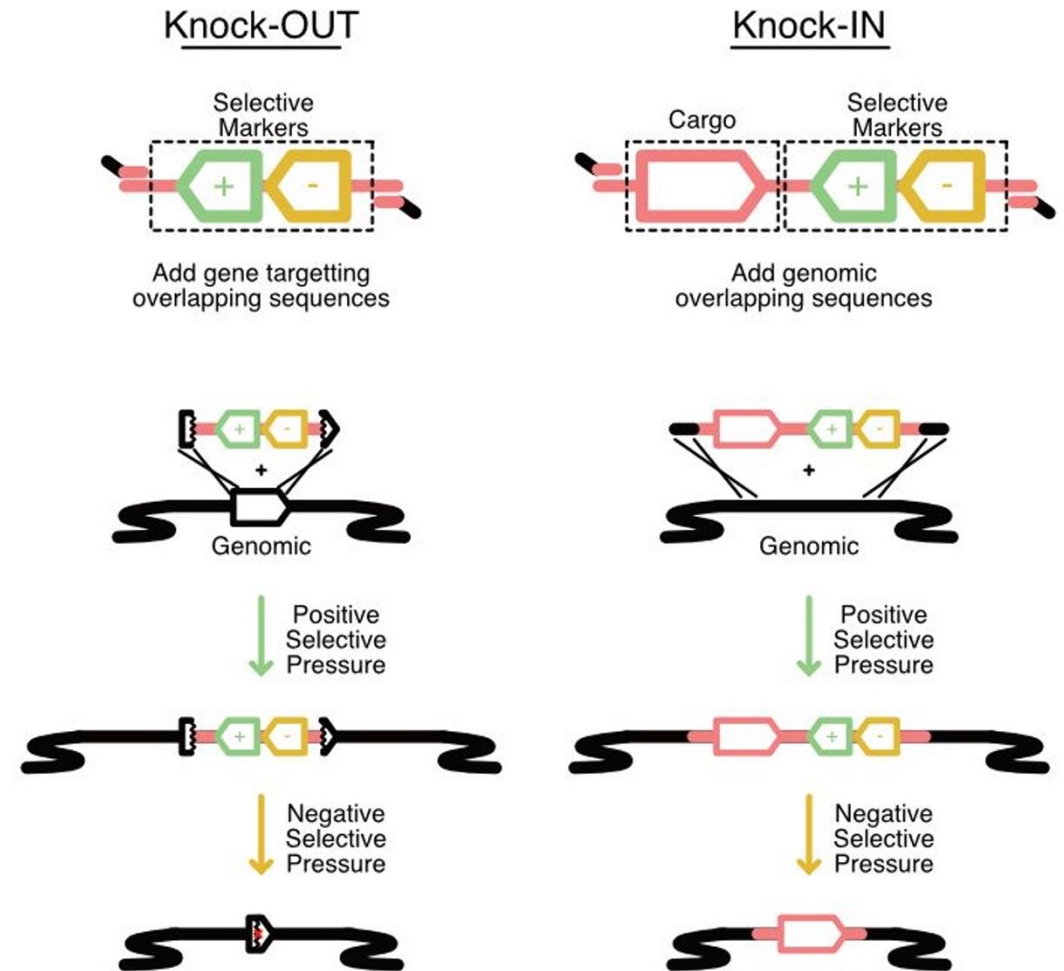


```
if(blue light):  
  inhibit(yf1)  
if(yf1):  
  activate(fixJ)  
if(fixJ-active):  
  express(cl)  
if(no-cl):  
  express(cdrAB)
```

# Engineering DNA: Editing genome by knocking IN or knocking-OUT



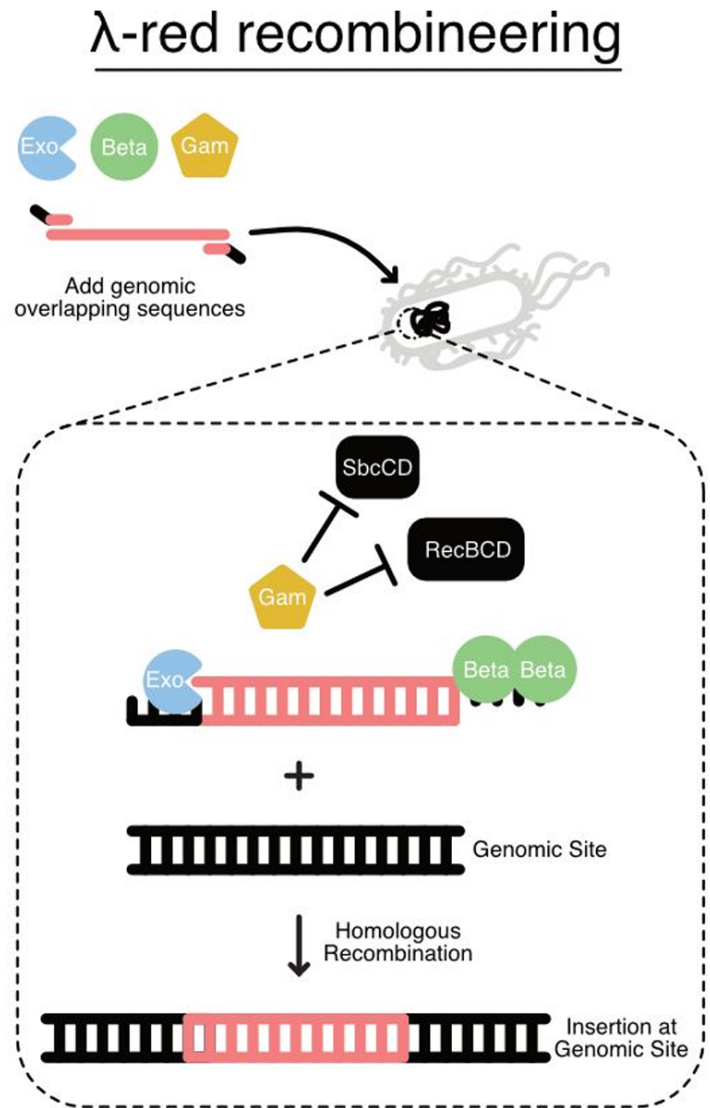
- Genomes can also be engineered
- One strategy for genome editing is to leverage homologous recombination to delete genes (Knock-OUT) or to introduce entirely new sequences (Knock-IN)
- This can be achieved by introducing selective markers to select for cells within populations that have either gained (positive selection) or lost the engineered DNA (negative selection)



# Engineering DNA: Using Recombineering to edit genomes



- Viral mechanisms for DNA integration have been co-opted to increase genomic engineering efficiency
- One approach, called 'recombineering', leverages the *E. coli* lambda prophage ([Yu D, et al., PNAS \(2000\), 97\(11\):5978-5983](#))
- Prophage proteins Exo, Beta, and Gam improve integration of DNA harboring genomic overlapping sequences by chewing back dsDNA to generate ssDNA, protecting ssDNA from degradation, and blocking host exonucleases, respectively

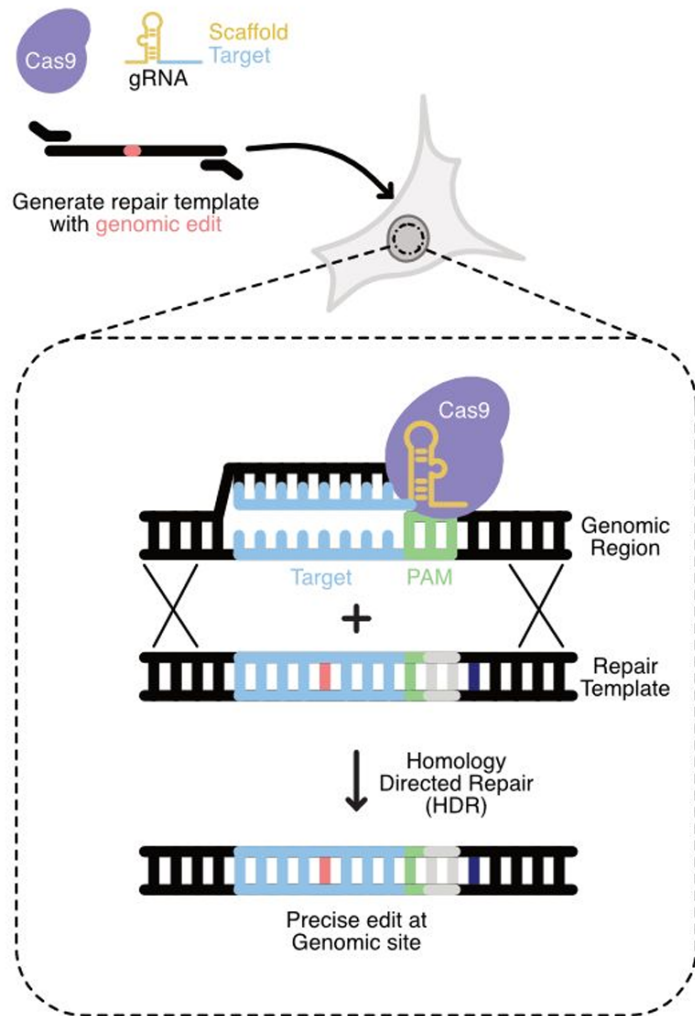


# Engineering DNA: Using CRISPR-Cas9 to make precise genome edits



- Adaptive-immunity mechanisms from bacteria have been co-opted to enable precise genomic edits
- Clustered Regularly Interspaced Short Palindromic (CRISPR) systems from diverse bacteria have enabled a revolution in genomic editing
- Cas9 is one of the most widely used CRISPR systems used for genome engineering ([Jinek M., et al., Science \(2012\) 337\(6096\):816-821](#))
- Cas9 is an RNA-guided endonuclease that makes double-stranded breaks at specific sites.
- Guide RNAs (gRNA) can be engineered target Cas9 to specific DNA sequences that precede PAM sites (NGG).
- One way to make genomic edits leverages homology directed repair to integrate an engineered repair template that eliminates the PAM site and subsequent retargeting.

## CRISPR editing



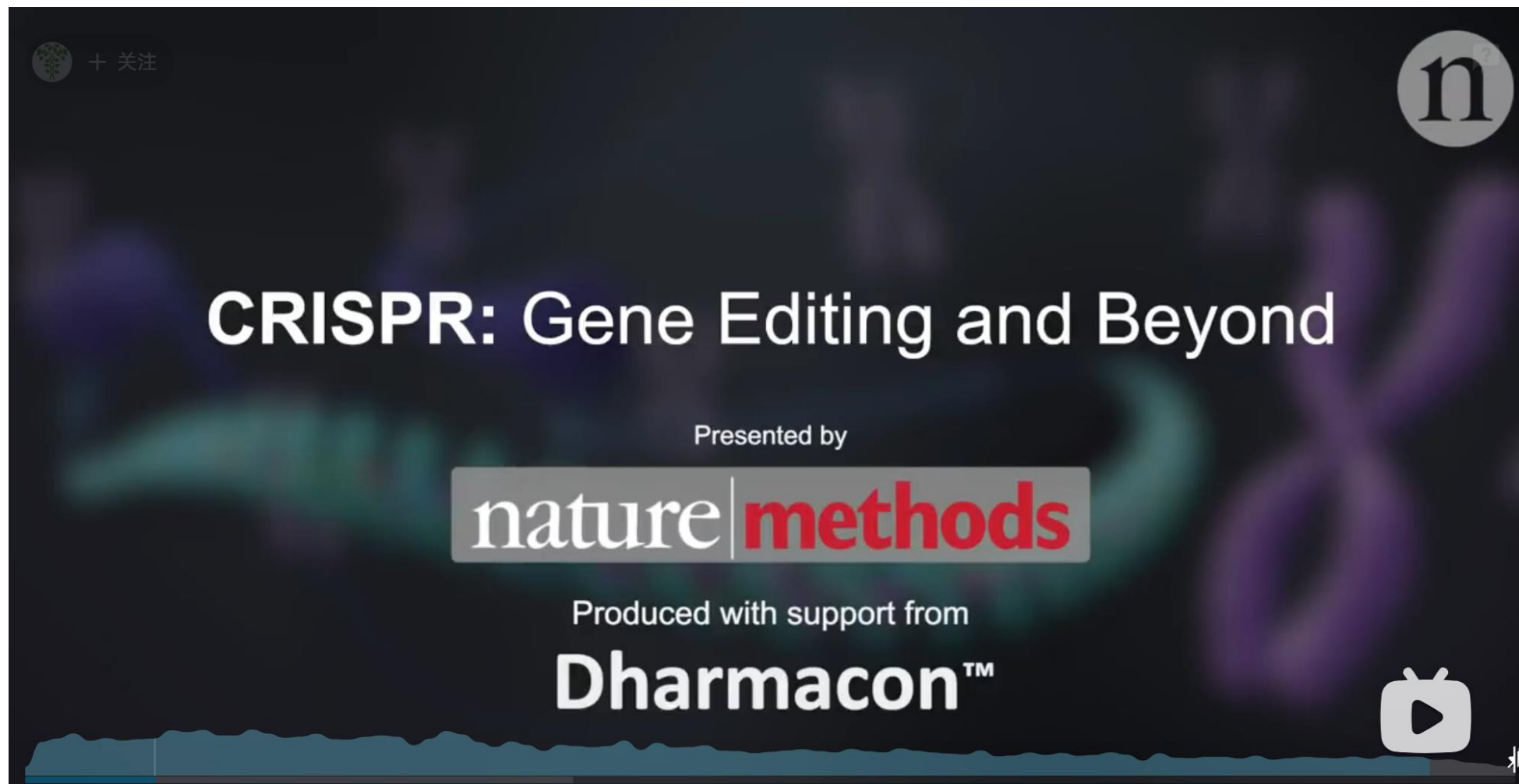




# CRISPR gene editing: mechanisms and applications



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[https://www.bilibili.com/video/BV15x411Z7kf/?spm\\_id\\_from=333.337.search-card.all.click&vd\\_source=4d96844e7fc7e94828f597310890ebb4](https://www.bilibili.com/video/BV15x411Z7kf/?spm_id_from=333.337.search-card.all.click&vd_source=4d96844e7fc7e94828f597310890ebb4)



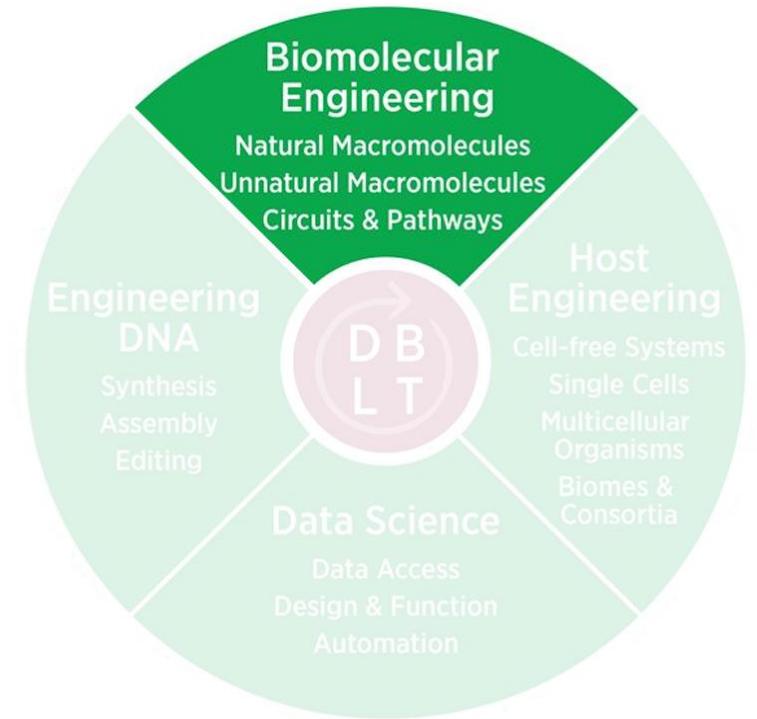


# Core tools for Biomolecular Engineering



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- What are our current tools for Biomolecular Engineering?
- Tools enable a Biomolecular Engineering DBTL
  - Comprehensive Mutagenesis libraries
  - Functional Screens
  - Functional Selections
  - Directed Evolution



Engineering Biology Research Consortium (2019). *Engineering Biology: A Research Roadmap for the Next-Generation Bioeconomy*. Retrieved from <http://roadmap.ebrc.org>. doi: 10.25498/E4159B.



# Biomolecular Engineering: Natural macromolecules

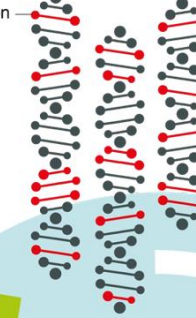
- Natural macromolecules including proteins and RNAs can be engineered to have new properties (e.g., thermostability) or perform new functions (e.g., binding, catalysis)
- Directed evolution can be used to tailor the function of proteins through rounds of sequence diversification, selection for function, and amplification of

## 1 Diversification

Random mutations are introduced in the gene for the enzyme that will be changed.



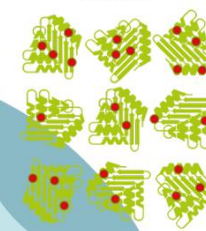
mutation



## 2 Selection

The genes are expressed in bacteria or *in vitro* to produce randomly mutated enzymes.

enzymes



mutation

The most active enzymes mutants are selected further

discarded enzyme

## 3 Amplification

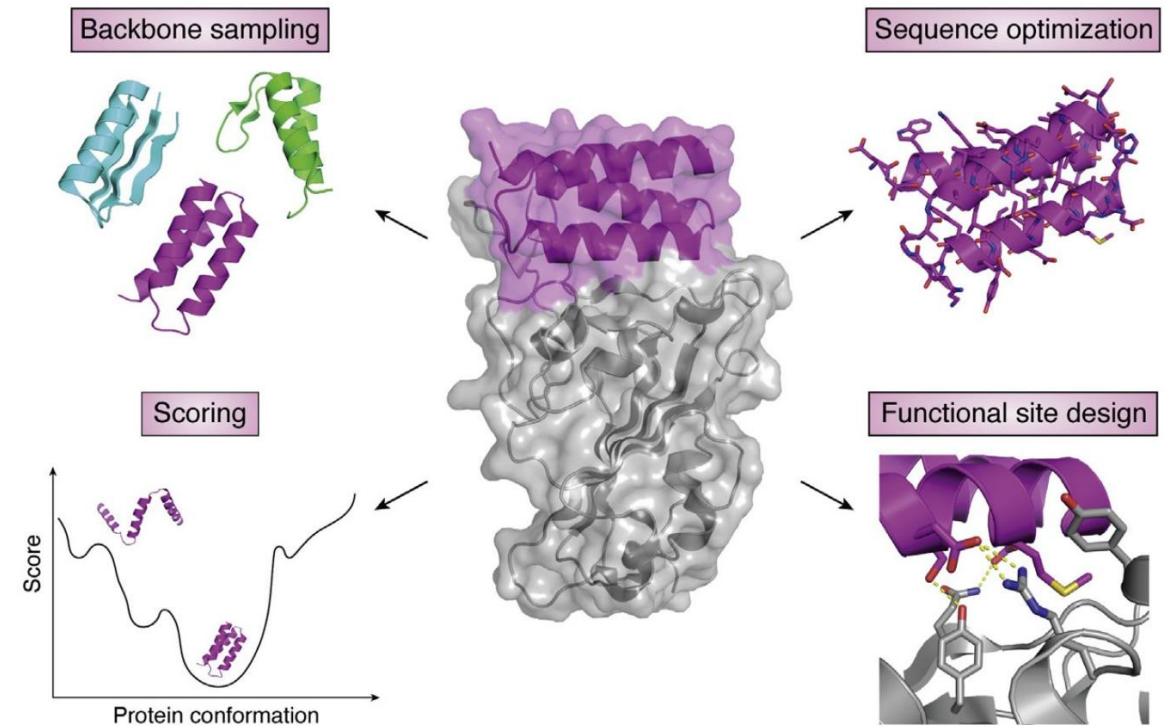
Active variants are amplified and new random mutations may be introduced at this stage

©Johan Jarnestad/The Royal Swedish Academy of Sciences

[The Nobel Prize.](#)

# Biomolecular Engineering: Unnatural macromolecules

- Entirely new protein sequences not seen in nature can be generated through *de novo* protein design
- *De novo* design enables exploration of protein sequence space beyond what nature has itself explored

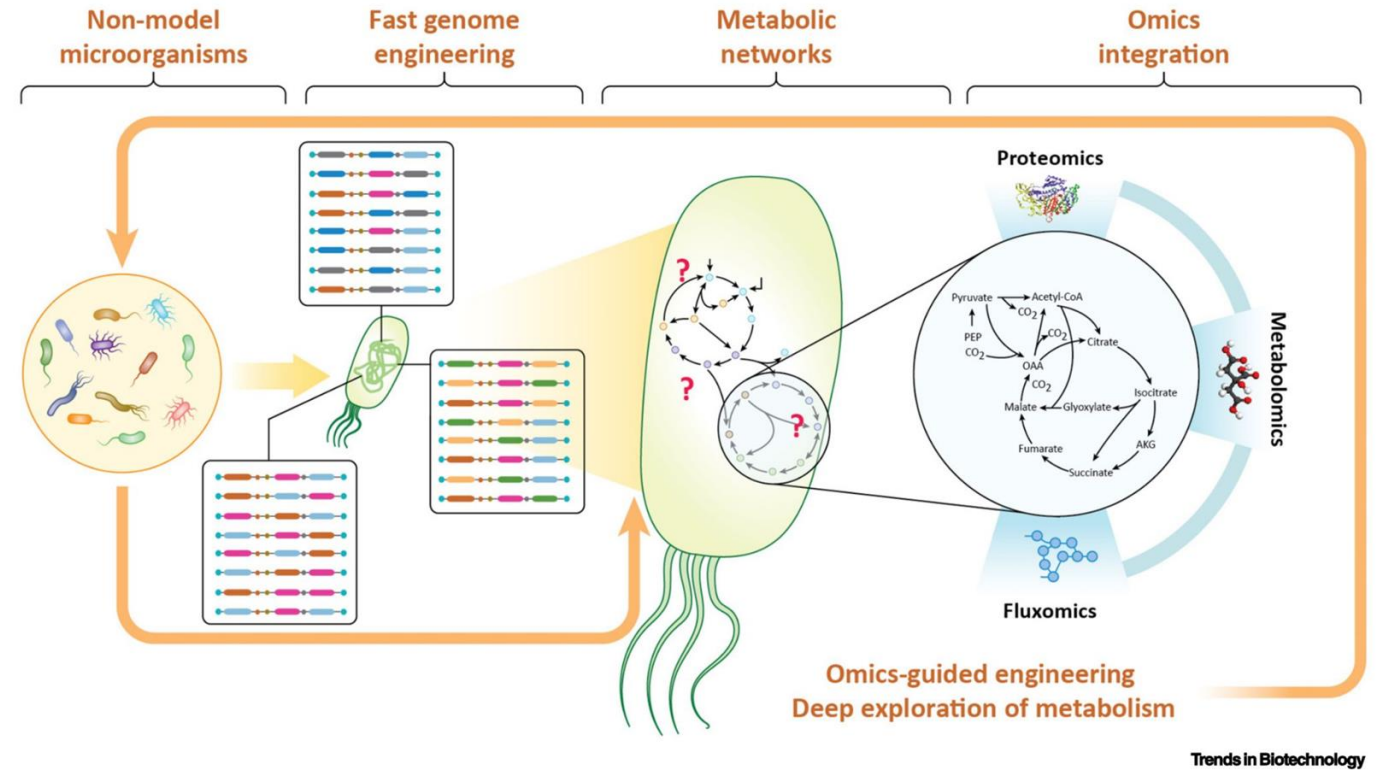


[Pan X. and Kortemme T. JBC Reviews \(2021\) 296 100558](#)



# Biomolecular Engineering: Circuits and Pathways

- In addition to individual biomolecules large scale networks of biomolecules
- Novel metabolic pathways can be constructed using retrosynthesis approaches
- Combinatorial libraries can be used to optimize expression of pathways proteins
- Adaptive laboratory evolution can be used to optimize the host genome for improved yield of a target molecule



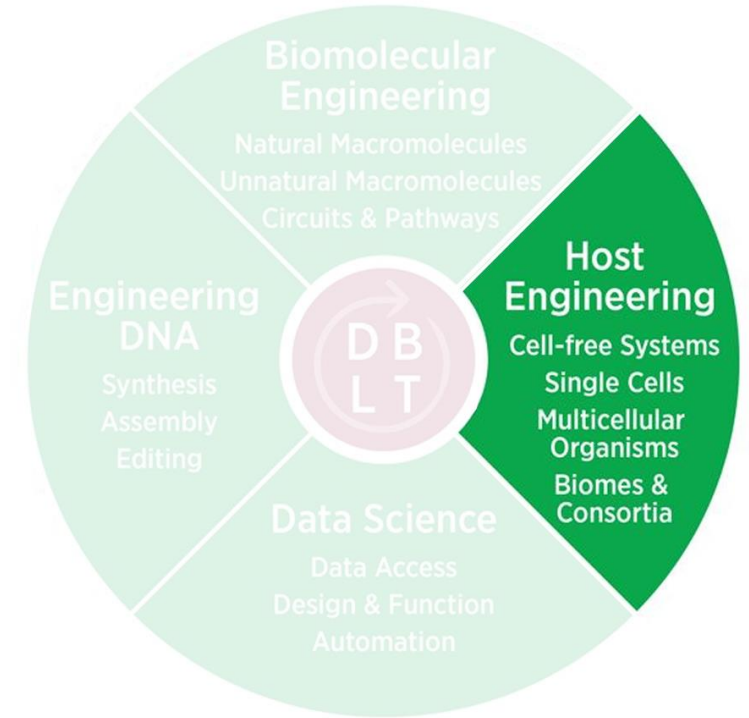
Trends in Biotechnology

[Gurdo N., Volke D.C., Nikel P.I. \(2022\) Trends in Biotechnology](#)



# Host engineering

- What types of hosts are there?
- Model & non-model systems
- Thinking across scales



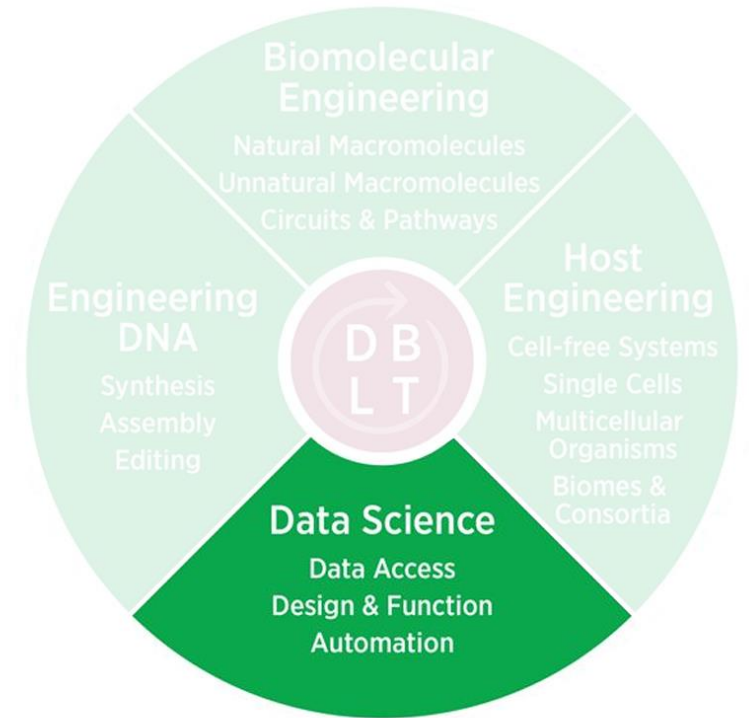
Engineering Biology Research Consortium (2019). *Engineering Biology: A Research Roadmap for the Next-Generation Bioeconomy*. Retrieved from <http://roadmap.ebrc.org>. doi: 10.25498/E4159B.

# Core tools for Data Science

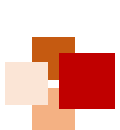


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- What data is out there?
- Modeling & Machine Learning
- Automation



Engineering Biology Research Consortium (2019). *Engineering Biology: A Research Roadmap for the Next-Generation Bioeconomy*. Retrieved from <http://roadmap.ebrc.org>. doi: 10.25498/E4159B.



# Data science: Omics data



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## Genome

Who is there?  
What can they make?



- Species identification
- Mutation rate in a population

## Transcriptome

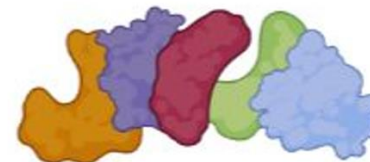
What are they  
saying to making?



- Gene regulation
- Population heterogeneity

## Proteome

What are they  
making?



- RNA regulation
- Protein stability & degradation

## Metabolome

What are they  
doing?



- Metabolic pathway efficiency
- Cellular inputs & outputs

Then integrate!



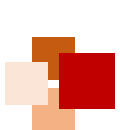


# ■ Data science: Databases



- National Center for Biotechnology Information (NCBI)
  - GenBank - NIH genetic seq. collection
  - Nucleotide BLAST - search gene seqs.
- UniProt
  - Protein seq. and functional information
- RCSB Protein Database (PDB)
  - Characterized protein structure and models
- Other, more specific DBs:
  - FPBase - fluorescent protein characterization and lineages





# Data science: Modeling genetic parts



- Can use first principles thermodynamics to predict function of parts in some cases
- Can integrate larger datasets with machine learning models to make more sophisticated predictions

1  
0 1 C  
0 1 0 1 0 1 T A G 0 T T C G A T  
0 1 0 1 1 1 0 A T 0 C 1 A C G C T G

## *De Novo DNA*

Predict RBS output based on both RBS and protein sequence



# Benchling

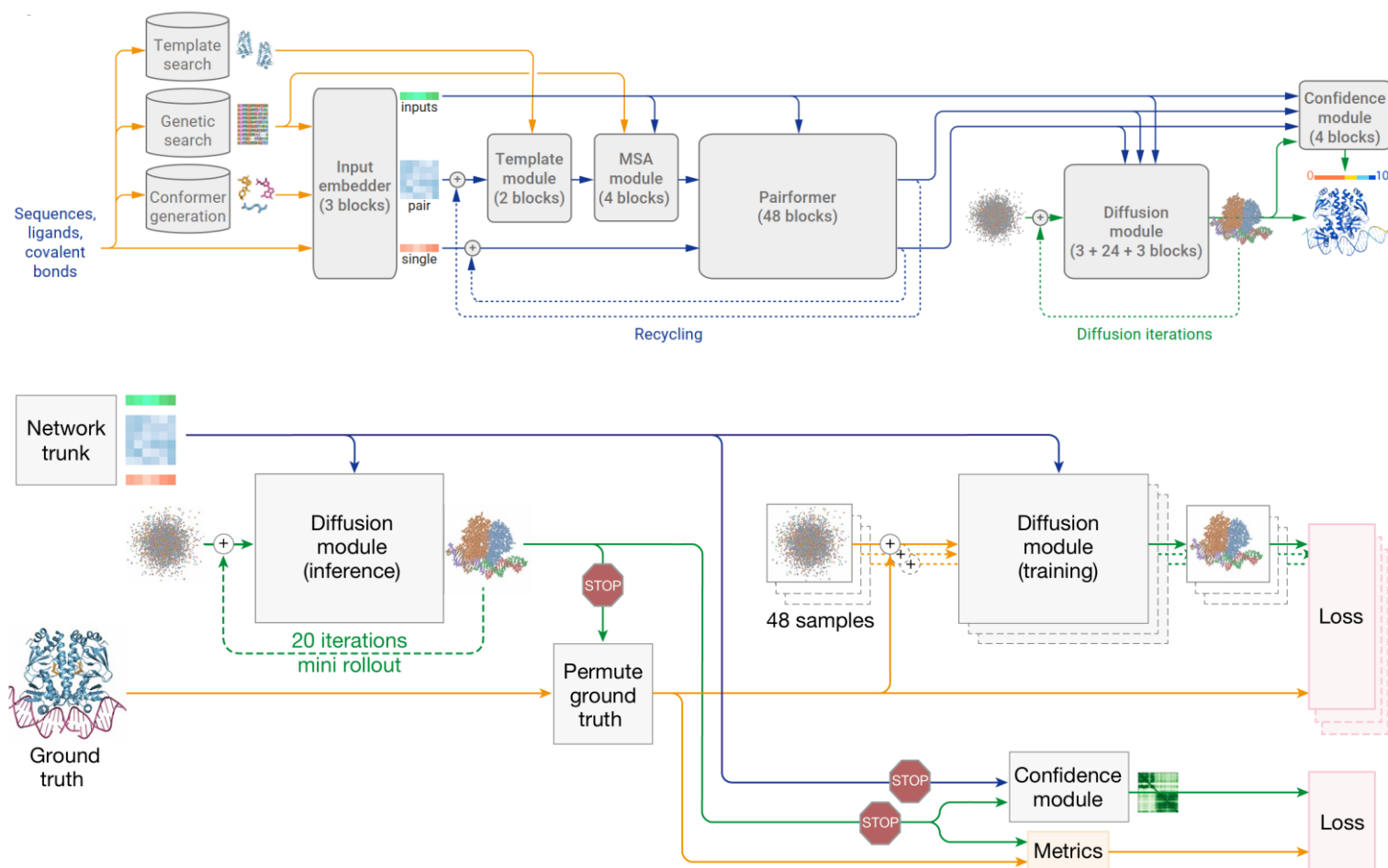
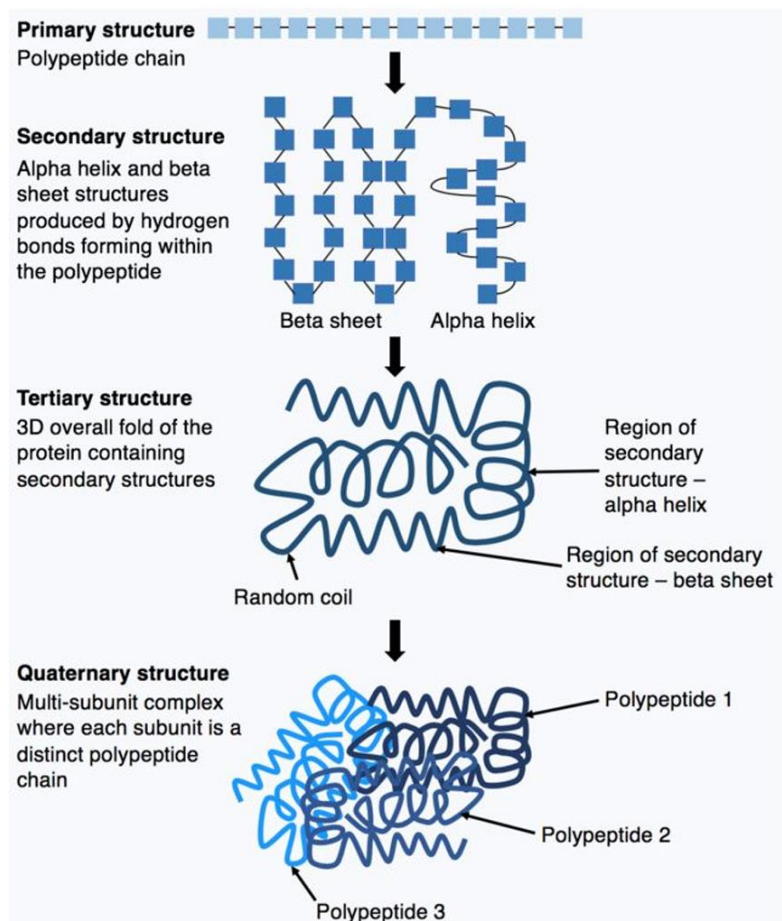
**RBS (Ribosome Binding Site)** is a sequence of nucleotides located upstream of the start codon (usually AUG) in the mRNA. It serves as a binding site for the ribosome, facilitating the initiation of protein synthesis.



# Data science: Modeling proteins (AlphaFold)

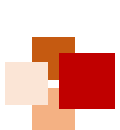


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Kep17, "Amino acid chains."

AlphaFold3, Google DeepMind (2024)

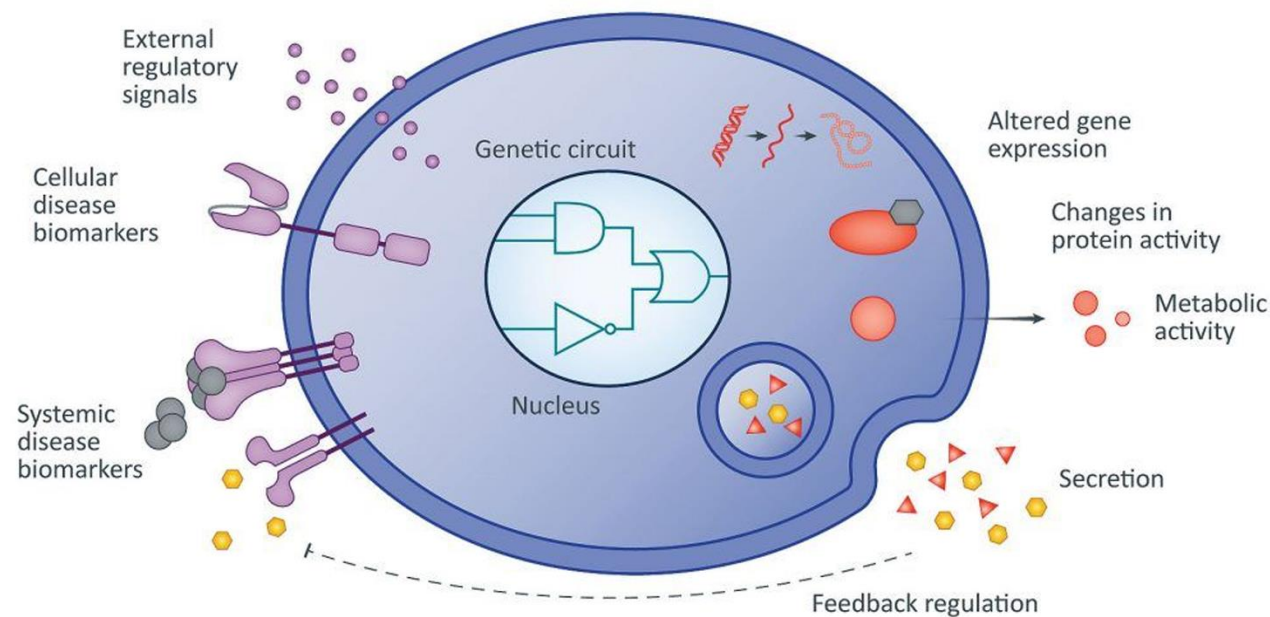


# Data science: Modeling whole systems



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- Quantify what is there (-omics)
- Determine inputs and outputs
- Apply equations to determine system functions - metabolic efficiency, protein activity, etc.
- NIH: 'Systems biology is an approach to understand the larger picture—be it at the level of the organism, tissue, or cell—by putting its pieces together.'



[Lu, T. \(2020\). Senti Biosciences.](#)



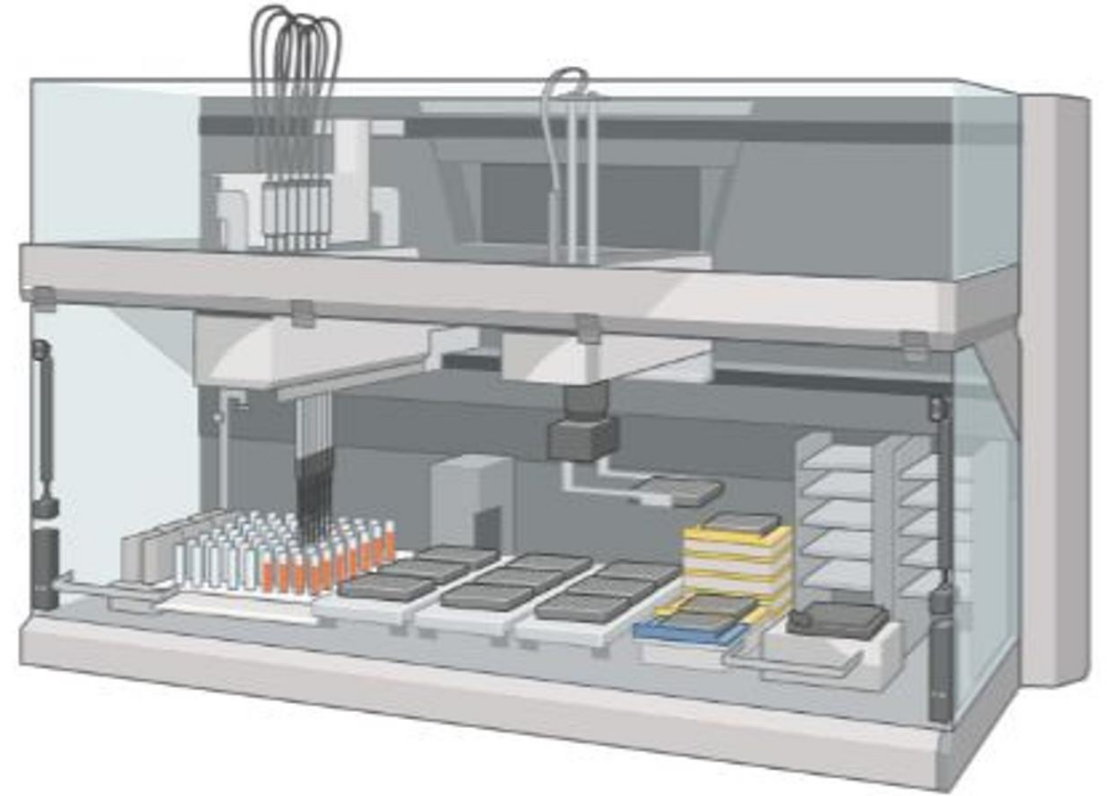




# Data science: Automation of laboratory processes



- Liquid handlers to automate Building & Testing genetic constructs
- Increase throughput to test multiple constructs simultaneously
- Integrated analysis for -omics





# References



- Dy AJ, Aurand ER, Friedman DC. YouTube resources for synthetic biology education. *Synthetic Biology*, 2019 Sep 5;4(1):ysz022. doi: 10.1093/synbio/ysz022.
- M. Eslami et al. Artificial Intelligence for Synthetic Biology, *Communications of the ACM*, May 2022, Vol. 65, No. 5, pp. 88-97.
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